SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 1

CANDIDATE NAME			
CLASS		**INDEX NUMBER**	

CHEMISTRY — **8872/02**
Preliminary Examination — **17 August 2012**
Paper 2 — **2 hours**

Additional Materials: Data Booklet
Writing Papers

Candidates answer Section A on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough work.

SECTION A:
Answer **all** questions in the space provided.

SECTION B:
Answer any **two** questions on separate answer paper.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
MCQ P1	
Section A	
B5	
B6	
B7	
Total	

This document consist of **17** printed pages and **1** blank page

Section A

Answer **all** the questions in this section in the spaces provided.

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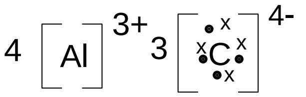
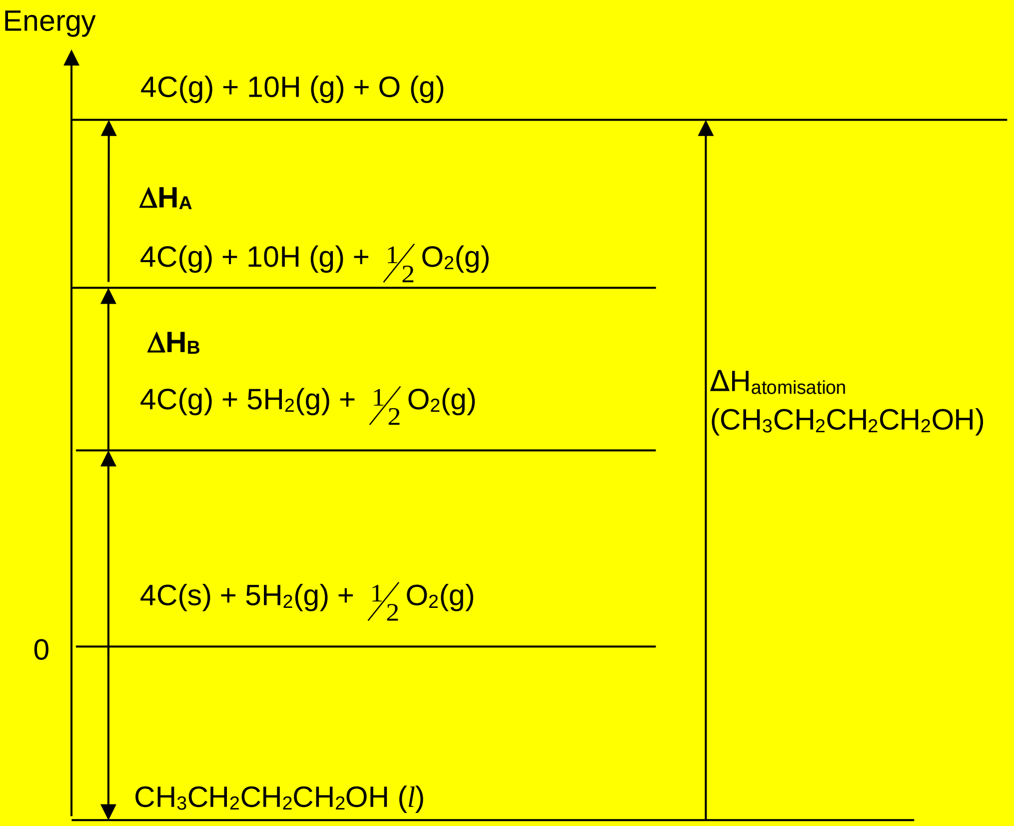
1	**(a)**		Aluminium is the third most abundant element (after oxygen and silicon) found in the Earth's crust. Due to its strong affinity to oxygen, it is almost never found in the elemental state. Aluminium reacts readily with carbon to form aluminium carbide, Al_4C_3, with relative formula mass of 144.0. It is stable up to 1400 °C and it decomposes in water to produce methane.
		(i)	Define the term *relative formula mass*.
			The relative formula mass of an ionic compound is defined as the ratio of the average mass of one formula unit of the compound to $\frac{1}{12}$ the mass of an atom of ^{12}**C** isotope.
		(ii)	State the electronic configuration of aluminium in aluminium carbide.
			$1s^2 2s^2 2p^6$
		(iii)	Draw the dot-and-cross diagram to illustrate the bonding in aluminium carbide.
			
		(iv)	Calculate the percentage by mass of carbon in aluminum carbide. **[4]**
			Percentage of C in Al_4C_3 = $\frac{3(12.0)}{4(27.0)+3(12.0)} \times 100\ \% = 25.0\ \%$
	(b)		Aluminium carbide can react with protic reagents such as hydrochloric acid to produce methane gas and aluminium(III) chloride. At high temperatures, the molecular mass of aluminium(III) chloride is half that measured under room conditions.
		(i)	State the molecular formula of aluminium(III) chloride at room temperature and suggest a reason for the difference in molecular mass of aluminium(III) chloride at high temperature and room temperature.
			Al_2Cl_6 At higher temperature there is **sufficient energy to overcome the dative bond** in aluminium(III) chloride thus the formula will become $AlCl_3$.
		(ii)	Write a balanced equation for the reaction between aluminium carbide and hydrochloric acid.
			$Al_4C_3(s) + 12HCl(aq) \longrightarrow 4AlCl_3(aq) + 3CH_4(g)$
		(iii)	Using the equation in **(b) (ii)**, calculate the maximum volume of methane gas that can be produced when 12.0 g of aluminium carbide reacts with 250 cm^3 of 2.0 mol dm^{-3} hydrochloric acid at standard temperature and pressure. **[6]**
			Amt of Al_4C_3 = $\frac{12.0}{4(27.0)+3(12.0)} = 0.08333$ mol Amt of HC*l* = 0.250 x 2.0 = 0.5 mol If Al_4C_3 is limiting, amt of HC*l* required = 12 x 0.08333= 1.000 mol So Al_4C_3 is in excess. HC*l* is limiting reagent Amt of CH_4 produced = $\frac{0.5}{12} \times 3 = 0.125$ mol Volume of CH_4 obtained = 0.125 x 22.4= 2.80 dm^3
			Total: [10]

2 (a) The standard enthalpy change of atomisation of a molecule is the energy absorbed when one mole of a gaseous molecule is atomised to form the gaseous atoms under standard conditions.

The atomisation reaction of butan-1-ol is:

$$CH_3CH_2CH_2CH_2OH\ (l) \rightarrow 4C\ (g) + 10H\ (g) + O\ (g)$$

An energy level diagram involving the atomisation of butan-1-ol is shown below:



(i) Define the term *standard enthalpy change of formation*.

Enthalpy change of formation is the **<u>energy change</u>** when **<u>one mole of compound is formed</u>** from its **<u>constituent elements under standard conditions.</u>**

(ii) Use the *Data Booklet* to obtain the values of the enthalpy change for **ΔH_A** and **ΔH_B**.

ΔH_B: **436 x 5 = 2180 kJ mol^{-1}**

ΔH_A: **496 x ½ = 248 kJ mol^{-1}**

(iii) Hence, using the information below and the values from **(a) (ii)**, calculate the enthalpy change of atomisation of butan-1-ol.

Given:
$\Delta H_{atomisation}$ (C) = 716 kJ mol^{-1}
$\Delta H_{formation}$ ($CH_3CH_2CH_2CH_2OH$) = -328 kJ mol^{-1}

ΔH_{atom} ($CH_3CH_2CH_2CH_2OH$)
= -(-328) + 4(716) + 2180 + 248
= **<u>+5620 kJ mol^{-1}</u>**

(iv) Calculate another value for the enthalpy change of atomisation of butan-1-ol using **only bond energies** values quoted from the *Data Booklet*.

ΔH_{atom} ($CH_3CH_2CH_2CH_2OH$)
= BE (C-O) + BE(O-H) + 3BE(C-C) + 9BE(C-H)
= 360 + 460 + 3(350) + 9(410)
= **<u>+5560 kJ mol^{-1}</u>**

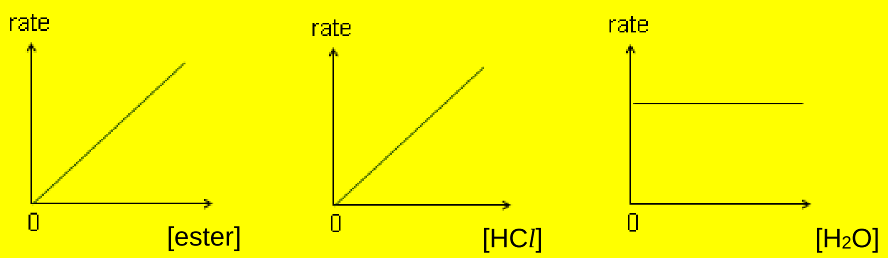
(v) Bond energy quoted from the *Data Booklet* are average values and may result in disparity in the answers in **(a) (iii)** and **(a) (iv)**. Suggest another reason for the difference in the values calculated in **(a) (iii)** and **(a) (iv)**. **[6]**

For **(a) (iv)**, vapourisation of $CH_3CH_2CH_2CH_2OH$ (*l*) into gaseous state is not taken into account.

(b) Butan-1-ol can be produced by the acidic hydrolysis of butyl ethanoate.

$$CH_3CH_2CH_2CH_2OCOCH_3 + H_2O \xrightarrow{HCl} CH_3CH_2CH_2CH_2OH + CH_3COOH$$

The following graphs were obtained when the rate and concentration of the reactants were investigated in the laboratory.



From the graphs, deduce the rate equation and hence determine the units for the rate constant. **[2]**

Rate = k[$CH_3CH_2CH_2CH_2OCOCH_3$][HC*l*]
mol^{-1} dm^3 s^{-1}

(c) The school laboratory has run out of alkaline iodine. Propose a series of reaction to differentiate a bottle of pentan-1-ol and pentan-2-ol. **[2]**

Step 1: (Elimination of H_2O from the alcohol)
Add conc H_2SO_4, 170°C
Step 2:
Add $KMnO_4$ in H_2SO_4 (aq), heat
Observation:
Purple $KMnO_4$ decolourises in both pentan-1-ol and pentan-2-ol but only effervescence is observed in the test tube containing penta-1-ol.

Total: [10]

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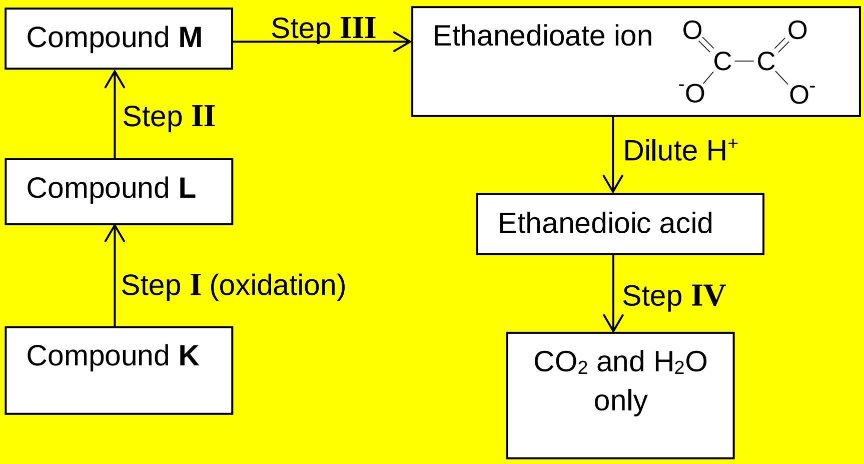
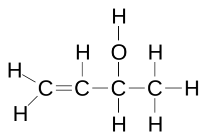
3 (a) The reaction scheme below shows how compound **K** can be converted into ethanedioic acid.

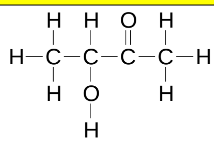
Compound **K** has molecular formula, C_4H_8O. It forms a yellow precipitate of CHI_3 when reacted with warm alkaline aqueous iodine. Compound **K** is also known to decolourise bromine water.

Compound **L** and **M** react with warm alkaline aqueous iodine to form a yellow precipitate. **M** evolves white fumes when reacted with phosphorus pentachloride but does not react with carbonate. On the other hand, only compound **L** decolourises bromine water.



(i) From the reaction scheme and the information provided, draw the displayed formulae of compounds **K** and **M**.

Compound **K**: 

Compound **M**: 

(ii) Suggest why potassium manganate(VII) cannot be used for step **I**.

Use of $KMnO_4$ will cause the **<u>cleavage of the carbon-carbon double bonds via oxidation</u>** which as a result **<u>L will not decolourise bromine water</u>** as mentioned in the question.

(not required by mark scheme: $K_2Cr_2O_7$ is used instead)
(do not accept if students write $K_2Cr_2O_7$ only as their answer)

(iii) State the reagents and conditions for steps **II** , **III** and **IV**.

[6]

Step **II**: **<u>H_2O, H_3PO_4, 65 atm, 300°C</u>**
Step **III**: **<u>I_2 in NaOH (aq), warm</u>** or alkaline aqueous iodine, warm
Step **IV**: **<u>$KMnO_4$, H_2SO_4 (aq), heat</u>**

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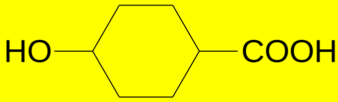
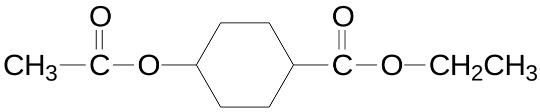
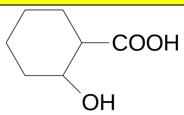
	(b)	**(i)**	Propose how ethanoic acid can be converted into ethanal.
			Step 1: Add $LiAlH_4$ in dry ether to ethanoic acid. Ethanoic acid will be reduced to ethanol Step 2: Add $K_2Cr_2O_7$ in H_2SO_4 (aq), distillation. Ethanol will be oxidised to ethanal.
		(ii)	Hot ethanoic acid followed by ethanol was reacted with compound **T** in the presence of concentrated H_2SO_4. Draw the structural formula of the organic product formed.  Compound **T**
			
		(iii)	There is an isomer of compound **T** which has a lower boiling point. Draw this isomer. **[4]**
			 Presence of intramolecular hydrogen bonding.
			Total: [10]

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4 The following table shows some properties of period 3 elements **V**, **W**, **X**, **Y** and **Z**.

Element	Boiling point / °C	Electrical conductivity	Reaction with oxygen
V	1410	0.10	No reaction
X	649	0.42	Burns very vigorously and brightly to form oxide
W	660	1.00	Solid formed reacts with acid and bases
Y	119	0	Forms a gaseous product
Z	44	0	Burns vigorously in excess oxygen

(a) Oxide of **X** and oxide of **Z** are dissolved in 2 separate beakers. The content in the beakers are then mixed. State the type of reaction that will occur. Explain your answer

[2]

Neutralisation.
Reason:
Oxides of X is basic in nature. It will form a basic solution when dissolved in water.
Oxides of Z is acidic in nature. It will form an acidic solution when dissolved in water.

(b) Explain the difference in boiling points of **V**, **X** and **Y**, with reference to their bonding and structure.

[3]

V has **giant molecular structure** with **strong covalent bonds between atoms**. It has the highest boiling point as the largest amount of energy is needed to overcome the strong covalent bonds.

X has **giant metallic lattice structure** with **strong electrostatic forces of attraction between cations and sea of delocalised electrons/ metallic bond** It has high boiling points as large amount of energy is needed to overcome the strong metallic bonds.

Y has **simple molecular structures** with **weak intermolecular van der Waal's forces of attraction**. It has lower boiling points because lesser energy is needed to overcome the weaker intermolecular van der Waals' forces of attraction.

(c) Write two chemical equations to show how the oxides of **X** and **W** reacts with hydrochloric acid.

$XO(s) + 2\ HCl(aq) \rightarrow XCl_2(aq) + H_2O\ (l)$

$W_2O_3(s) + 6\ HCl(aq) \rightarrow 2\ WCl_3(aq) + 3\ H_2O\ (l)$

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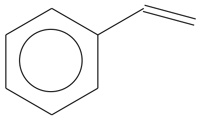
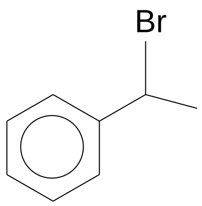
	(d)	Describe the reactions, if any, of the chlorides of element **W** with water, suggesting the pH of the resulting solution and writing equations, where appropriate.
		[3]
		W^{3+} undergoes **hydrolysis** as it has **high charge density** and is able to **polarise** and weaken O-H bond in **water to release H^+** and give an **acidic** solution. WCl_3 (s) + 6 H_2O (*l*) → $[W(H_2O)_6]^{3+}$ (aq) + 3 Cl^- (aq) $[W(H_2O)_6]^{3+}$(aq) → $[W(H_2O)_5(OH)]^{2+}$ (aq) + H^+ (aq) pH of solution = **3**
		Total: [10]

Section B
Choose 2 questions to answer

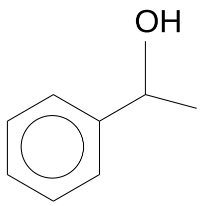
5	**(a)**	15 cm^3 of hydrocarbon **J** was exploded in excess oxygen and allowed to cool to room temperature. There was an overall reduction of 45 cm^3. After the gas mixture was passed through aqueous sodium hydroxide a further contraction of 120 cm^3 was observed. **J** reacts with gaseous hydrogen bromide to form **K**. Aqueous sodium hydroxide was added to **K** to produce **L**. **L** decolourises aqueous potassium dichromate(VI) to give **M**. Both **L** and **M** give a yellow precipitate with aqueous alkaline iodine. **K** reacts with aqueous sodium cyanide to produce **N**, which forms **P** upon reaction with aqueous hydrochloric acid.
	(i)	Using the combustion data given, determine the molecular formula of **J**. **[3]**
		Volume of CO_2 = 120 cm^3 Initial total volume – final total volume = 45 (vol of **J** +vol of unreacted O_2+vol of reacted O_2)–(vol of CO_2+vol of unreacted O_2) = 45 15 + vol of reacted O_2 – 120 = 45 Vol of reacted O_2 = 150 $\frac{x}{1} = \frac{120}{15}$ x = 8 $\frac{x+\frac{y}{4}}{1} = \frac{150}{15}$ y = 8 **<u>J is C_8H_8</u>**.
	(ii)	Using your answer in **(a) (i)** deduce, with reasoning, the structures of compounds **J, L, M** and **P**. **[9]**
		(1) J undergoes **<u>electrophilic addition with HBr(g).</u>** ➔ **J** contains **<u>C=C</u>**. **(2) K** undergoes **<u>nucleophilic substitution</u>** with **<u>aqueous NaOH</u>** to form **L**. ➔ **K** is a **<u>halogenoalkane</u>**. **(3) L** undergoes **<u>oxidation</u>** with **<u>$K_2Cr_2O_7$</u>** to form **M**. ➔ **L** may be a **<u>1^0 or 2^0 alcohol</u>**. ➔ **M** may be an **<u>acid or ketone.</u>** **(4) M** undergo **<u>oxidation</u>** with aqueous **<u>alkaline iodine</u>**. ➔ **M** contains $R-\overset{\overset{\Large O}{\Vert}}{C}-CH_3$ ➔ **L** is a 2° alcohol. **(5) L** undergoes **<u>nucleophilic substitution</u>** to form **N**. ➔ **N** is a **<u>nitrile</u>**. **(6) N** undergoes **<u>acidic hydrolysis</u>** to form **P**. ➔ **P** is a **<u>carboxylic acid</u>**.



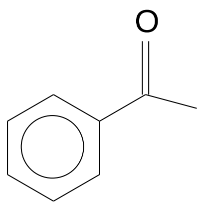
J



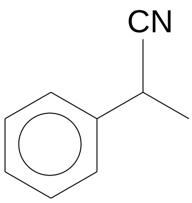
K



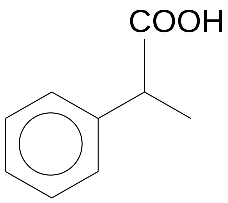
L



M



N



P

(b) Esters can be synthesized from the reaction of a carboxylic acid and an alcohol. The equation for the reaction is shown below:

$$RCOOH\ (l) + R'OH\ (l) \rightleftharpoons RCOOR'\ (l) + H_2O\ (l)$$

In an experiment, 1 dm^3 of 2.5 mol dm^{-3} of acid is reacted with 1 dm^3 of 1.5 mol dm^{-3} of alcohol. The equilibrium constant, K_c of this reaction is 1.6.

(i) Define *dynamic equilibrium*.

Dynamic Equilibrium is one where **rate of forward reaction equals to rate of backward reaction** for a **reversible reaction**.

(ii) Write an expression for the equilibrium constant, K_c. Calculate the concentration of the ester formed at equilibrium. **[4]**

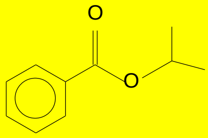
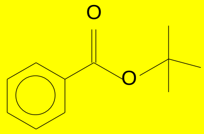
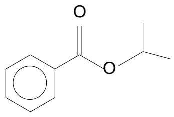
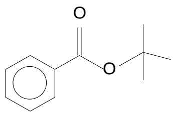
$$Kc = \frac{[RCOOR'][H_2O]}{[RCOOH][R'OH]}$$

	RCOOH (*l*)	+ R'OH (*l*) $\rightleftharpoons$	RCOOR' (*l*)	+ H_2O (*l*)
Initial conc	**1.25**	**0.75**	**0**	**0**
Change	-x	-x	+x	+x
Equilibrium con	1.25 – x	0.75 - x	x	x

$1.6 = \frac{x^2}{(1.25-x)(0.75-x)}$

$0.6x^2 - 3.2x + 1.5 = 0$

x = **0.52 $moldm^{-3}$** or x = 4.8 (rejected)

	(c)		Esters are commonly used in food flavouring. Describe a chemical test to distinguish the following esters  and . **[2]**
			Test: Add **<u>$KMnO_4$ in dilute H_2SO_4</u>** to each compound separately and **<u>heat</u>**. Observation:  : **<u>Decolourisation of purple</u>** $KMnO_4$.  : **<u>No decolourisation</u>** of purple $KMnO_4$.
	(d)		On reaction with ethanolic silver nitrate, bromocyclohexane forms more precipitate as compared to chlorocyclohexane while no precipitation was observed for fluorocyclohexane. Explain why these observations were made. **[2]**
			C-X Bond strength: C-F > C-C*l* > C-Br Ease of cleavage of bond: C-F < C-C*l* < C-Br Thus, more ppt is observed for bromocyclohexane as compared to chlorocyclohexane. No ppt is observed as C-F bond is too strong for nucleophilic substitution to occur.
			Total: [20]

6	**(a)**	Most of the Earth's crust consists of solid oxides. They were formed when elements develop an oxide coating as a result of oxidation in air. Aluminium foil, often used in food preparation, develops a thin oxide layer over the aluminium metal to protect it from corrosion.	
		(i)	The oxides of magnesium, silicon and phosphorus differ in their reactions with water. Account, with reference to the structure and bonding of the oxides, for the above reactions and state the approximate pH of the solutions formed. Write equations where appropriate. **[5]**
			Magnesium oxide is a **<u>giant ionic compound with strong ionic bonds</u>**. Due to its high lattice energy, it **<u>reacts slowly</u>** with water to produce a weakly alkaline solution. **$MgO + H_2O \rightleftharpoons Mg(OH)_2$ pH = 9** SiO_2 has a **<u>giant molecular structure with strong covalent bonds between</u>** Si and O **<u>atoms</u>**. Large amount of energy is required to break these strong covalent bonds, hence, SiO_2 is **<u>insoluble in water</u>**. P_4O_6 (or P_4O_{10}) has a **<u>simple molecular structure with weak intermolecular van der Waals' forces of attraction</u>**. P_4O_6 (or P_4O_{10}) reacts with water to form acidic solution. **Either $P_4O_6 + 6\ H_2O \rightarrow 4\ H_3PO_3$ or $P_4O_{10} + 6H_2O \rightarrow 4\ H_3PO_4$ pH = 2 or 3**
		(ii)	A student attempted to remove the oxide layer on the aluminium foil. He placed the aluminium foil strips into a polystyrene cup and added 60 cm^3 of 1.2 mol dm^{-3} sulfuric acid. 3.0 g of the aluminium oxide coating the metal reacted based on the following reaction. $Al_2O_3 + 3H_2SO_4 \rightarrow Al_2(SO_4)_3 + 3H_2O$ The maximum temperature change was recorded to be 11.5 °C. Given that the specific heat capacity of the solution = 4.2 J g^{-1} K^{-1} and assuming the process to be 75% efficient, calculate the standard enthalpy change of neutralisation for the reaction.
			Amount of H_2SO_4 = 60/1000 x 1.2 = 0.07200 mol Amout of Al_2O_3= 3 /102 = 0.0294 mol H_2SO_4 is the limiting reagent. Amount of H_2O = 0.07200 mol Q' = mcΔT = 60 x 4.2 x 11.5 = 2898 J Q = 100/75 x 2898 = 3864 J $\Delta H_n^\theta = -\dfrac{Q}{n_{H_2O}}$ = - 3864 / 0.07200 = - 53666 J mol^{-1} = **<u>- 53.7 kJ mol^{-1}</u>**

		(iii)	Another student proposed a method to remove the aluminium oxide layer using aqueous sodium hydroxide instead. With the aid of balanced chemical equation(s), comment on whether you agree with this proposal.
			Agreed, NaOH may be used as Al_2O_3 is an amphoteric oxide (Al^{3+} has high charge density to polarise the large O anionic cloud, resulting in covalent character.) $\mathbf{Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2Na[Al(OH)_4]}$
		(iv)	With reference to your answer in **(a)(ii)**, predict if the standard enthalpy change of neutralisation between ethanoic acid and aluminium oxide is less exothermic or more exothermic. **[12]**
			Since ethanoic acid is a weak acid, **some energy is absorbed to dissociate the H^+ ions**. The enthalpy change of neutralisation would be **less negative than that in (ii) / less exothermic.**
	(b)		Compound **W** is found to contain 75.3% carbon, 10.6% hydrogen and 14.1% oxygen by mass. The relative molecular mass of **W** is 112. When **W** is oxidised with acidified potassium manganate(VII), a compound **X**, $C_4H_6O_3$, is produced together with propanoic acid. No white fumes are evolved when **W** was mixed with phosphorus pentachloride. **X** produces a pale yellow precipitate when warmed with alkaline solution of iodine, accompanied by a solution which, when treated with dilute hydrochloric acid, gave compound **Y**. **Y** may also be obtained when $NC – CH_2 – CN$ is heated with a dilute acid. On heating with concentrated sulphuric acid, 2 mole of **X** reacted with 1 mole of ethane-1,2-diol to produce a sweet smelling compound **Z**. Deduce the structure of compounds **W, X, Y** and **Z** and explain the chemistry of the reactions described above. **[8]**

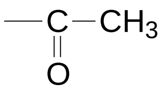
	C	H	O
% by mass	75.3	10.6	14.1
Amt	75.3/12 = 6.275	10.6/1 = 10.6	14.1/16 = 0.881
mol ratio	7	12	1

n($C_7H_{12}O$) = 112 $\Rightarrow$ n = 1
W has a molecular formula: **<u>$C_7H_{12}O$</u>**

W undergoes oxidation with acidified $KMnO_4$ to produce **X** and propanoic acid.
W is likely to contain **<u>C=C</u>** since 2 products are formed. (not likely to be ester as there is only one oxygen)

W does not undergo **<u>nucleophilic substitution</u>** with PCl_5.
W <u>does not contain alcohol</u> functional group

X undergoes **<u>oxidation</u>** with alkaline aq. Iodine.
Since **X** is a product of oxidation, it cannot contain alcohol group. Thus, it contains

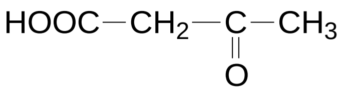


Solution formed undergoes **<u>acidification</u>** with HC*l* (aq) to give **Y**.
Solution **is <u>carboxylate salt</u>** and **Y** is **<u>acid</u>.**

NC – CH_2 – CN undergoes **<u>acidic hydrolysis</u>** with dilute acid to form **Y**.

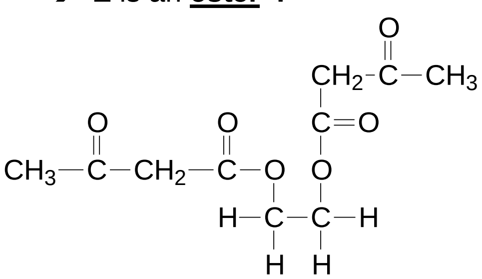
➔ **Y** is $HO_2C – CH_2 – CO_2H$

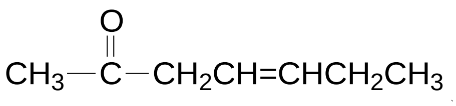
➔ **X** is



X reacts with ethane-1,2-diol via **<u>esterification / nucleophilic substitution</u>** to produce **Z.**

➔ **Z** is an **<u>ester</u>** **:**



➔ **W:** 

Total: [20]

7	An acidic buffer consists of a *weak acid*, such as a carboxylic acid, and the salt of its conjugate base. Within a certain pH range, the acidity of a carboxylic acid is important and affects the buffering capacity of the buffer solution. The acid dissociation constant of the carboxylic acid is directly proportional to its buffer capacity.	
	(a)	Define the term *weak acid*. **[1]**
		A *weak acid* is one that **<u>dissociates partially</u>** in water to produce H^+.
	(b)	25.0 cm^3 of citric acid, HA, with a pH of 3.1 undergoes neutralisation with 25.60 cm^3 of 0.50 mol dm^{-3} of potassium hydroxide. Support your answer with relevant argument why citric acid is a weak acid. **[2]**
		Amt of KOH = 0.0128 mol Amt of citric acid HA = 0.0128 mol [citric acid] = 0.512 mol dm^{-3} $[H^+] = 10^{-3.1} = 7.94 \times 10^{-4}$ mol dm^{-3}] Since $[H^+]$ is not equal to [acid] ➔ Citric acid is a weak acid.
	(c)	With the aid of equations, explain how citric acid and its sodium salt function as a buffer solution. **[2]**
		Let citric acid be HA and its sodium salt be NaA When small amount of acid is added, $A^- + H^+ \rightarrow HA$ The added H^+ is removed as A^- When small amount of alkaline is added: $HA + OH^- \rightarrow A^- + H_2O$ For both cases pH remains fairly constant.
	(d)	Rank the following carboxylic acids based on increasing pH: 2-chloropropanoic acid, 2-fluoropropanoic acid, 2-methylpropanoic acid Explain your answer. **[4]**
		2-methylpropanoic acid has an **<u>electron-donating alkyl</u>** group that **<u>intensifies the negative charge on the O atom</u>** of the carboxylate ion. 2-chloropropanoic acid and 2-fluoropropanoic acid have **<u>electron-withdrawing</u>** substituents that **<u>disperse the negative charge on the O atom</u>** of the carboxylate ions. **<u>F is more electronegative than C*l*</u>** and disperses the negative charge on the O atom of the carboxylate ion to a **<u>greater extent</u>**. Stability of carboxylate ion: 2-methylpropanoic acid < 2-chloropropanoic acid < 2-fluoropropanoic acid pH: 2-methylpropanoic acid > 2-chloropropanoic acid > 2-fluoropropanoic acid

(e) An unknown ester produces 2-methylpropanoic acid as the only product when it was subjected to the following reactions:

1. Heating with dilute sulfuric acid followed by
2. Heating with aqueous potassium manganate(VII)

In an investigation, the following kinetics data were obtained for Step 1.

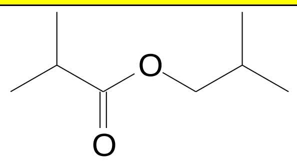
With 1.0 mol dm^{-3} sulfuric acid:

Time / min	0	2	4	6	8	10
[**ester**] / mol dm^{-3}	0.0100	0.0068	0.0047	0.0032	0.0021	0.0015

With 2.0 mol dm^{-3} sulfuric acid:

Time / min	0	2	4	6	8	10
[**ester**] / mol dm^{-3}	0.0100	0.0046	0.0021	0.0010	0.0004	0.0002

(i) Draw the structural formula of the ester.



(ii) Using the same axes, plot graphs of [ester] against time for:

(a) Reaction with 1 mol dm^{-3} of sulfuric acid.
(b) Reaction with 2 mol dm^{-3} of sulfuric acid.

Hence, find the rate equation showing your working clearly.

Graph
- Axes labels, units and scales
- Labelled graphs
- Correctly plotted points
- Smooth curves drawn

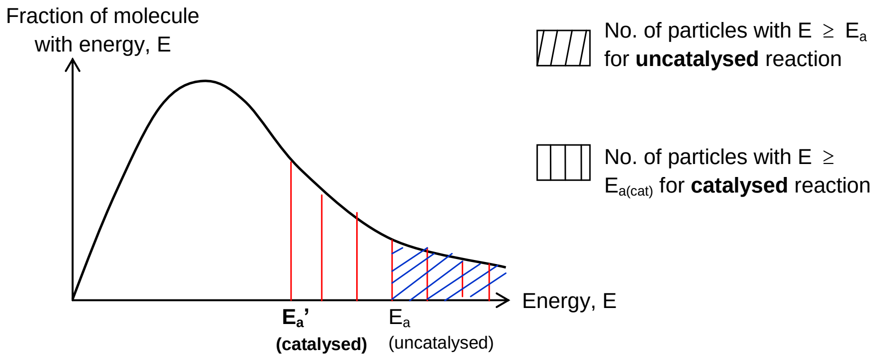
From the graph, the reaction with [H_2SO_4] = 2.0 mol dm^{-3} has a **constant half-life of 1.8 min**. Hence **order wrt H_2SO_4** is **one**
(Shows constant half-life of 1.8 min on graph)

For [H^+] = 1.0 mol dm^{-3},
Initial rate = 0.0025 mol dm^{-3} min^{-1}

For [H^+] = 2.0 mol dm^{-3},
Initial rate = 0.0050 mol dm^{-3} min^{-1}

When **[H^+] is doubled** while [**X**] is constant, **rate doubles**. **Order wrt X** is **one**
(Shows tangents at t = 0, calculates initial rates correctly and gives an appropriate explanation)

Rate = k[X][H^+]

		(iii)	In the reverse reaction, where esterification takes place between 2-methylpropanoic acid and an alcohol, the addition of a small amount of concentrated sulfuric acid increases the rate of reaction significantly. With the help of suitable diagram, explain how concentrated sulfuric acid affects the rate of reaction. Suggest if the addition of concentrated sulfuric acid affects the equilibrium constant. **[11]**
			 Diagram - Axes labels - Curve starts from origin - Correctly labelled activation energies and shadings for catalysed and uncatalysed reactions Deduct 1 mark for every mistake. **Number of molecules with E ≥ E_a increases**, **number of effective collisions increases**. Since **rate is proportional to frequency of effective collisions**, Hence, the rate of reaction increases. Concentrated sulfuric acid **does not affect the equilibrium constant**.
			Total: [20]

END